QUESTION 1 [29]

1.1. A geology company in Polokwane requires your expertise since you are doing Computer Science at the University of Limpopo. The company needs a system that can help its geologists make a comprehensive report when they do their work. What kind of a system can you recommend? Which components should form part of that kind of system? List the shortcomings that exist. By implementing a Geographic information system (GIS), the geology company in Polokwane can significantly improve its data management, analysis, and reporting processes, leading to better decision-making and increased productivity. (09)

1.2. Which law is represented by the following expression? (04)

$\neg(A \wedge B) \equiv (\neg A \vee \neg B)$

1.3. Suppose P and Q are two propositions. Which of the following expressions are equivalent? (06)

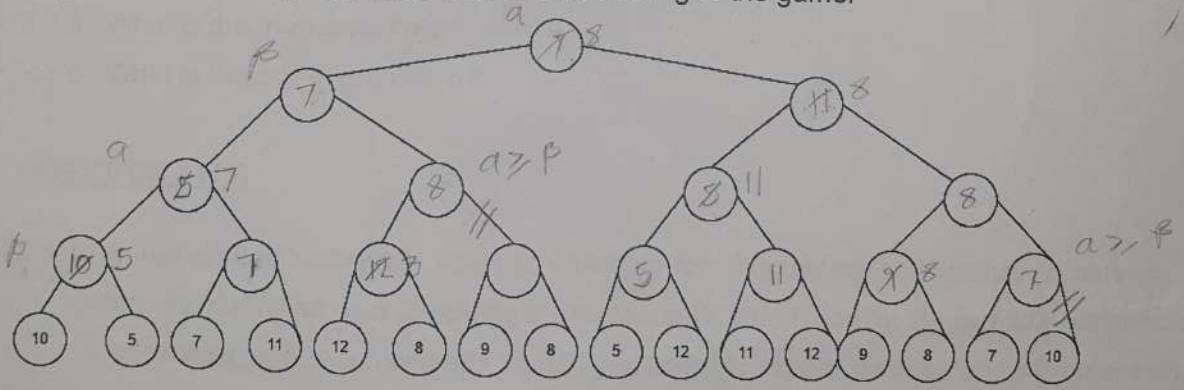
i. $\neg(\neg P \wedge Q)$.

ii. $(P \wedge Q) \vee (P \wedge \neg Q) \vee (\neg P \wedge Q)$

iii. $(P \vee \neg Q)$

iv. $(P \wedge Q) \vee (P \wedge \neg Q) \vee (\neg P \wedge \neg Q)$

1.4. Use the alpha-beta pruning mechanism to calculate the solution that can lead to winning the game and show the path that leads to the winning of the game. (10)



QUESTION 2 [20]

2.1. Use the following expressions to draw a semantic network graph. (10)

John is a cat.

John caught a bird.

Peter owns John.

John is black in colour.

Cats like milk.

The cat sat on the mat.

A cat is a mammal.

A bird is an animal.

All mammals are animals.

Mammals have fur.

John is a cat

John caught a bird

Peter owns John

2.2. Use Figure 2.1 below to answer the following questions.

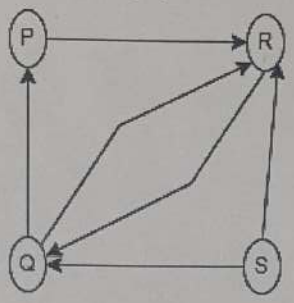


Figure 2.1

2.2.1 Whose house(s) has Q visited? (01)

2.2.2 Is the graph strongly connected? Explain your answer. (02)

2.2.3 Create the transitive closure matrix. (04)

2.2.4 Which node(s) has visited the most nodes? (01)

2.2.5 What is the in-degree for S? (01)

2.2.6 What is the out-degree for S? (01)

QUESTION 3 [11]

3.1. A well-crafted restaurant script is essential for maintaining consistency in service, ensuring customer satisfaction, and upholding the restaurant's brand identity. By carefully crafting a script for the restaurant, which components should you cover to create a cohesive and professional dining experience for your customers? (05)

3.2. Which slots of the frame are represented by the following statements.

3.2.1 The "hotel phone" might be a special instance of "phone," which might be an instance of a "communication device." (03)

3.2.2 An important feature of frames is the ability to attach procedural instructions to a slot. (03)